WORKED SOLUTION

STEP II 2012, Question 10

Mechanics • Friction and Moments

STEP II 2012, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q10

The key idea

The rod is a chord of the circular cross-section, so every normal reaction points straight at the centre O and the angle ϕ between the rod and each radius is fixed by the geometry: $\cos\phi = a/r$. The only way the rod can move is by turning about O, so both ends slip together, and we know which way friction acts at each end. Taking moments about the *midpoint* kills the weight and gives the first relation at once; moments about O and one resolution then give $\tan\theta$.

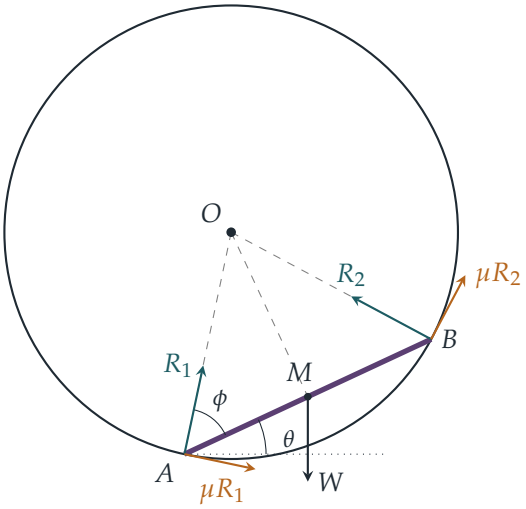
Solution

Setting up

Model the rod as a uniform rigid rod of weight W, touching the rough cylinder only at its ends A (lower) and B (higher). Let O be the centre of the circular cross-section and M the midpoint of AB. Since $OA = OB = r$, the triangle OAB is isosceles, so OM is perpendicular to AB and

$$\cos\phi = \frac{a}{r}, \qquad \sin\phi = \frac{\sqrt{r^2 - a^2}}{r}, \qquad OM = \sqrt{r^2 - a^2} = r\sin\phi,$$

with $0 < \phi < \frac{\pi}{2}$. The angle between the rod and the radius is ϕ at *both* ends.



Ask yourself Which way does friction act at each end?

The ends stay on the circle and AB stays the same length, so OM keeps its length and stays perpendicular to the rod: the only possible motion is a rotation about O. Since $\theta \neq 0$, M is not directly below O, and the weight has a moment about O turning the rod towards the horizontal (in the diagram, clockwise: A would move up the wall and B down it). The reactions pass through O, so only friction can resist this. Both ends would move together, so on the point of slipping friction is limiting at both: μR_1 and μR_2 act along the tangents at A and B, both in the anticlockwise sense about O, as shown.

The relation between R_1 and R_2

Take moments about M, which removes the weight. The line of action of R_1 is AO, which makes angle ϕ with AM, so its perpendicular distance from M is $a\sin\phi$. The friction at A acts along the tangent, which is perpendicular to OA, so its distance from M is $a\cos\phi$. The same holds at B. In the diagram R_1 turns the rod clockwise about M, while R_2 and both friction forces turn it anticlockwise, so

$$R_1\, a\sin\phi = R_2\, a\sin\phi + \mu R_1\, a\cos\phi + \mu R_2\, a\cos\phi.$$

Dividing by $a\cos\phi$ and rearranging,

$$\mu(R_1 + R_2) = (R_1 - R_2)\tan\phi,$$

as required. In particular $R_1 > R_2$: the lower end presses harder.

Finding $\tan\theta$

Moments about O. The reactions pass through O, and each friction force has moment arm r. The weight acts at M, and since OM is perpendicular to the rod it makes angle

θ with the vertical, so the horizontal distance of M from O is $OM\sin\theta = r\sin\phi\sin\theta$. Hence

$$\mu(R_1+R_2)\,r = Wr\sin\phi\sin\theta \quad\Longrightarrow\quad \mu(R_1+R_2) = W\sin\phi\sin\theta. \tag{1}$$

Resolving perpendicular to the rod (towards the side containing O). Each reaction makes angle ϕ with the rod, so contributes $R_i\sin\phi$. Each friction force makes angle ϕ with this perpendicular direction; at A it points away from O's side of the rod and at B towards it. The weight contributes $-W\cos\theta$. So

$$(R_1+R_2)\sin\phi - \mu(R_1-R_2)\cos\phi = W\cos\theta. \tag{2}$$

Write $S = R_1+R_2$. The first result gives $R_1-R_2 = \mu S\cot\phi$, so (2) becomes

$$S\,\frac{\sin^2\phi - \mu^2\cos^2\phi}{\sin\phi} = W\cos\theta,$$

while (1) says $\dfrac{\mu S}{\sin\phi} = W\sin\theta$. Dividing (the rod is inclined, not vertical, so $0<\theta<\frac{\pi}{2}$ and $\cos\theta\neq 0$),

$$\tan\theta = \frac{\mu}{\sin^2\phi - \mu^2\cos^2\phi}.$$

Multiply top and bottom by r^2 and use $r^2\sin^2\phi = r^2-a^2$, $r^2\cos^2\phi = a^2$:

$$\tan\theta = \frac{\mu r^2}{r^2 - a^2 - \mu^2 a^2} = \frac{\mu r^2}{r^2 - a^2(1+\mu^2)},$$

as required.

Deducing $\lambda < \phi$

Ask yourself Where does an inequality come from, when all we have is an equation?

From the signs. The rod is inclined at θ to the horizontal with $\theta\neq 0$, so $0<\theta<\frac{\pi}{2}$ and $\tan\theta>0$. As $\mu>0$, the numerator μr^2 is positive, so the denominator must be positive too:

$$r^2\sin^2\phi - \mu^2 r^2\cos^2\phi > 0 \quad\Longleftrightarrow\quad \tan^2\phi > \mu^2 = \tan^2\lambda.$$

Both ϕ and λ lie in $(0,\frac{\pi}{2})$, where tan is positive and increasing, so $\tan\phi > \tan\lambda$ gives $\lambda<\phi$.

Checks

- Smooth limit: as $\mu\to 0$, $\tan\theta\to 0$, so a smooth rod can only rest horizontally, with M directly below O. Correct.

- Numbers: with $r = 5$, $a = 3$, $\mu = \frac{1}{2}$ we get $\tan\phi = \frac{4}{3} > \frac{1}{2}$ and $\tan\theta = \frac{12.5}{25 - 11.25} = \frac{10}{11}$. Solving the two force equations for R_1, R_2 in vector form at this angle (and at thousands of random r, a, μ) makes the moment about O vanish and satisfies $\mu(R_1 + R_2) = (R_1 - R_2)\tan\phi$ to within 10^{-12}.
- Physical sense of $\lambda < \phi$: λ is the angle of friction. If $\lambda \geq \phi$, friction is strong enough to hold the rod at every inclination, so it is never on the point of slipping and no such θ exists.

Common slips

- Taking the reactions perpendicular to the *rod* rather than along the radii: the contact is with the cylinder, so the normal is the cylinder's normal.
- Putting the two friction forces in opposite directions along the rod. Both ends move in the same rotational sense about O, so friction is anticlockwise about O at both.
- Resolving along and perpendicular to the rod for the first result: both equations contain W. Moments about M avoid it.

Beyond STEP

The deduction $\lambda < \phi$ is the friction-cone idea: the total reaction at a rough contact must lie within angle λ of the normal. Here the rod is in contact at two points, and the problem is about whether the two friction cones leave room for the three forces to meet at a point. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body statics more fully, including problems where the direction of impending motion has to be worked out.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.